

CS & IT ENGINEERING



Operating System

CPU Scheduling

Lecture -3

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Recap of Previous Lecture



Topic

SRTF Scheduling

Topic

LF Scheduling

Topic

LRTF Scheduling

Topics to be Covered



Topic

HRRN Scheduling

Topic

Priority Based Scheduling

Topic

Round Robin Scheduling



Topic : HRRN (Highest Response Ratio Next)



Objective: Not only favors short jobs but decreases the WT of longer jobs.





Topic : HRRN (Highest Response Ratio Next)

Scheduling Criteria: *Response Ratio / Tie breaker $\Rightarrow$ SJF*

Type of Algorithm: *Non-preemptive*

$$\text{Response Ratio} = \frac{W + S}{S}$$

W = Wait Time

S = Service/Burst Time

$\Rightarrow$ no starvation
 $\Rightarrow$ not practical because
BT of processes are not known
in advance



Topic : HRRN (Highest Response Ratio Next)

Process	Arrival Time	Burst Time
P1	0	3
P2	2	6
P3	4	4
P4	6	5
P5	8	2

At Time 9 :-

$$RR(P3) = \frac{5+4}{4} = 2.25 \text{ (Highest)}$$

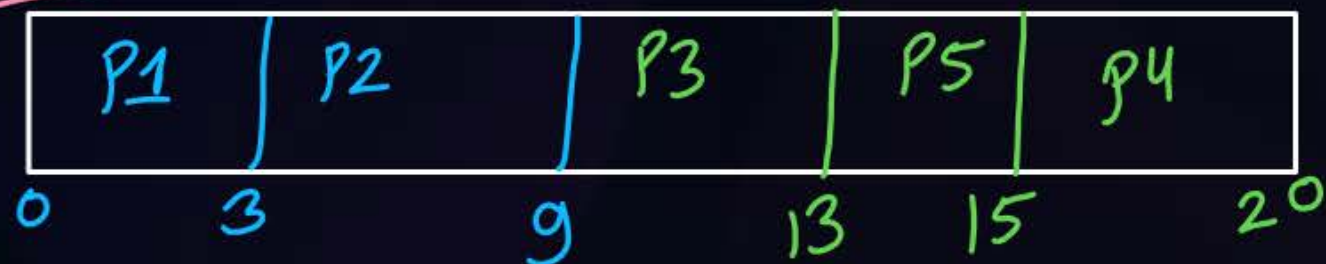
$$RR(P4) = \frac{3+5}{5} = 1.6$$

$$RR(P5) = \frac{1+2}{2} = 1.5$$

SJF :-



HRRN :-



At time 13 :-

$$RR(P4) = \frac{7+5}{5} = 2.4$$

$$RR(P5) = \frac{5+2}{2} = 3.5 \text{ (Highest)}$$

Ques)

	AT	BT
P1	0	5
P2	1	3
P3	2	2
P4	4	1

<u>P1</u>	P3	P4	P2	
0	5	7	8	11

At Time 5 :-

$$RR(P2) = \frac{4+3}{3} = 2.33$$

$$RR(P3) = \frac{3+2}{2} = 2.5 \text{ (Highest)}$$

$$RR(P4) = \frac{1+1}{1} = 2$$

At time 7 :-

$$RR(P2) = \frac{6+3}{3} = 3$$

$$RR(P4) = \frac{3+1}{1} = 4 \text{ (Highest)}$$



Topic : Priority Based Algorithm

Scheduling Criteria: *highest priority process first* | Tie breaker $\Rightarrow$ given in Questⁿ
(FCFS by default)

Type of Algorithm: *Preemptive*
or
non-preemptive





Topic : Priority Based Algorithm

Non-preemptive



Process	Arrival Time	Burst Time	Priority
P1	0	4	4
P2	1	2	5
P3	2	3	6
P4	3	1	10(Highest)
P5	4	2	9
P6	5	6	7

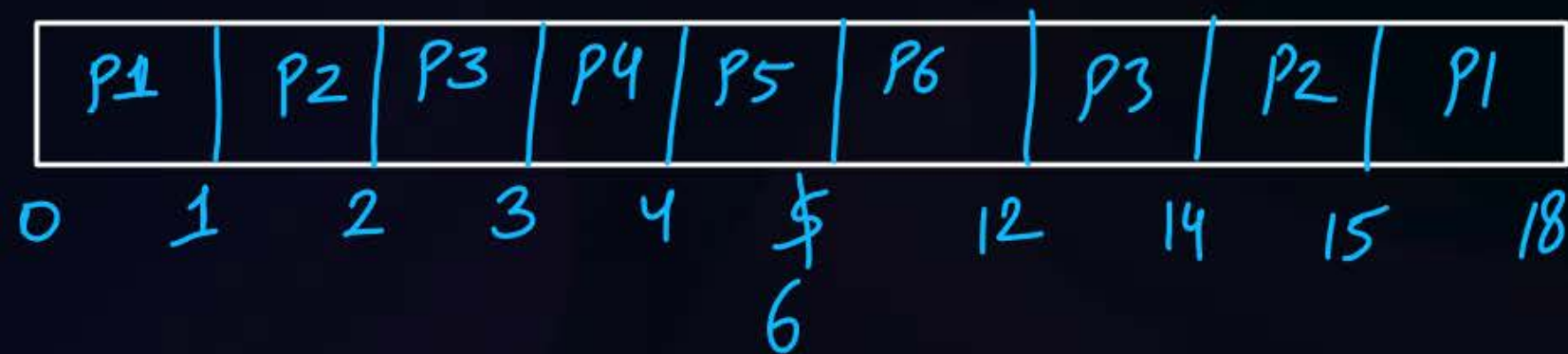
P1	P4	P5	P6	P3	P2	
0	4	5	7	13	16	18



Topic : Priority Based Algorithm

Preemptive

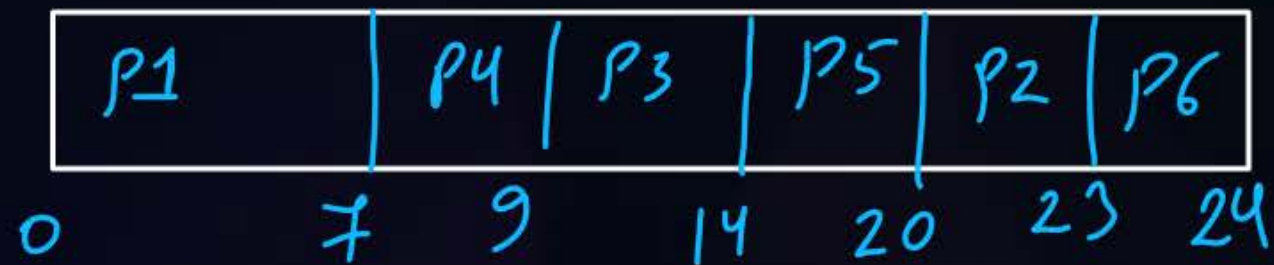
Process	Arrival Time	Burst Time	Priority
P1	0	4	4
P2	1	2	5
P3	2	3	6
P4	3	1	10(Highest)
P5	4	2	9
P6	5	6	7





Topic : Priority Based Algorithm Question Non-Preemptive

Process	Arrival Time	Burst Time	Priority
P1	0	7	9
P2	1	3	4
P3	2	5	2
P4	3	2	1 (Highest)
P5	4	6	3
P6	5	1	8





Topic : Priority Based Algorithm Question Preemptive

Process	Arrival Time	Burst Time	Priority	CT	TAT	WT
P1	0	7	9	24	24	17
P2	1	3	4	17	16	13
P3	2	5	2	9	7	2
P4	3	2	1 (Highest)	5	2	0
P5	4	6	3	15	11	5
P6	5	1	8	18	13	12

P1	P2	P3	P4	P3	P5	P2	P6	P1	
0	1	2	3	5	9	15	17	18	24

$$\text{avg WT} = \frac{49}{6} = 8.17$$



Topic : Priority Based Algorithm

Advantages:

1. Better response for real time situations

Disadvantages:

2. Low Priority Processes may suffer from starvation

⇒ solution of starvation ⇒ Aging

possible only when dynamic
priority of processes possible

increase priority of processes by
one when they wait for a certain
amount of time

↓
After a long waiting process's priority will
become highest.



Topic : Round Robin (RR)

Scheduling Criteria: *FCFS + Quantum*

Type of Algorithm: *Preemptive*

Quantum (time slice) $\Rightarrow$ A max duration for which a process runs on CPU at a time.

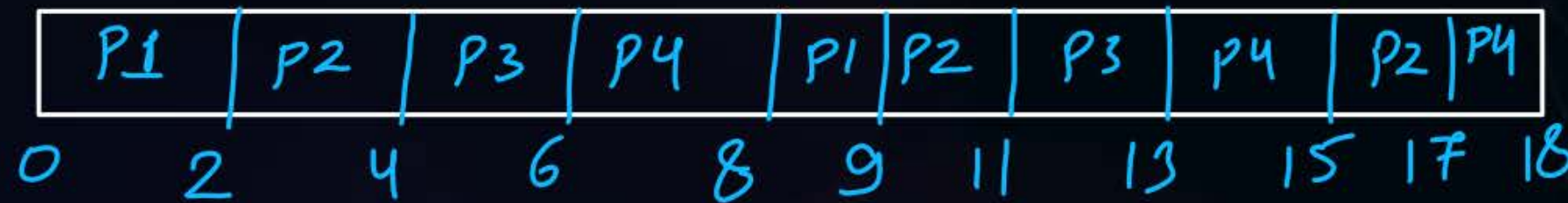


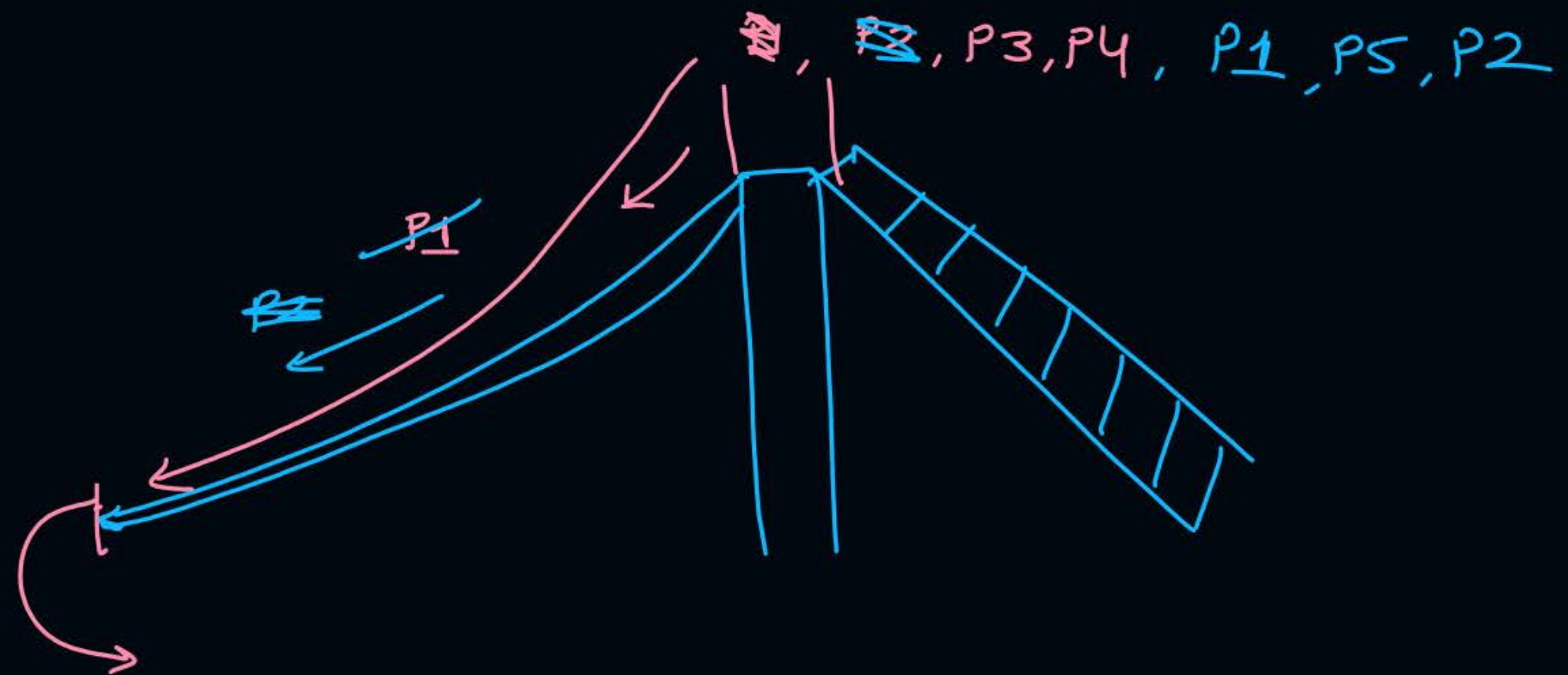
Topic : Round Robin (RR)

$Q = 2$

Process	Arrival Time	Burst Time
P1	0	3
P2	0	6
P3	0	4
P4	0	5

Ready Queue
P1, P2, P3, P4







Topic : Round Robin (RR)

$Q = 2$

Process	Arrival Time	Burst Time
P1	0	5
P2	1	4
P3	2	1
P4	3	3

Time	Front Ready Queue	Rear
0	P1	
2	P2 , P3, P1	
4	P3 , P1, P4, P2	
5	P1, P4, P2	

p1	p2	p3	p1	p4	p2	p1	p4	
0	2	4	5	7	9	11	12	13



Topic : Round Robin (RR)

$Q = 2$

Process	Arrival Time	Burst Time
P1	0	6 $\frac{6}{2} = 3$
P2	1	5 $\lceil \frac{5}{2} \rceil = 3$
P3	2	4 $\frac{4}{2} = 2$
P4	3	3 $\lceil \frac{3}{2} \rceil = 2$
P5	4	2 $\frac{2}{2} = 1$
P6	5	4 $\frac{4}{2} = 2$

Time	Ready Queue
0	P1
2	P2 , P3, P1
4	P3 , P1, P4, P5, P2
6	P1, P4, P5, P2, P6, P3 X X X X X X

P1	P2	P3	P1	P4	P5	P2	P6	P3	P1	P4	P2	P6	
0	2	4	6	8	10	12	14	16	18	20	21	22	24

for n processes

$$\text{no. of context switches} = \left(\sum_{i=1}^n \left\lceil \frac{BT_i}{Q} \right\rceil \right) - 1$$



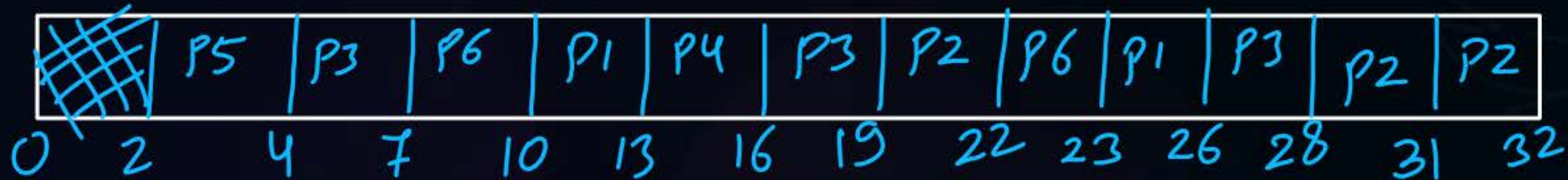
Topic : Round Robin (RR)

$Q = 3$

Process	Arrival Time	Burst Time
P1	5	6
P2	8	7
P3	3	8
P4	6	3
P5	2	2
P6	4	4

Time	Ready Queue
0	
2	P5
4	P3 , P6
7	P6 , P1, P4, P3
10	P1, P4, P3, P2, P6 x x x x

no. of context switches
= 11





Topic : Round Robin (RR)

$Q = 3$

Process	Arrival Time	Burst Time
P1	0	12
P2	0	5
P3	3	9
P4	5	6
P5	2	8
P6	4	2
P7	1	7

H. W.





2 mins Summary

Topic

HRRN Scheduling

Topic

Priority Based Scheduling

Topic

Round Robin Scheduling



Happy Learning

THANK - YOU

